

MA262 ALGEBRA & COMPLEX NUMBERS SUPPLEMENT

CONTENTS

1. Complex Numbers	1
1.1. Operations with Complex Numbers	2
2. Exponentials of Complex Numbers	4
2.1. Exponential functions and their derivatives	6
2.2. More About Conjugates	7
2.3. The polar representation	7
3. The Factorization of Polynomials	9
3.1. Polynomials of degree two	9
3.2. Polynomials of degree three	11
3.3. Polynomials of degree four	12
3.4. Polynomials of higher degree	12
Appendix A. The Roots of Unity	14
Appendix B. More About Complex Exponential Functions	16

The material covered in these notes is some times taught in high school precalculus courses, but this is often not the case and we will need to use it in MA262. These notes will not be covered in class, but students may ask questions about them during office hours. This material will not be used until the middle of the semester, when we get to Lesson 23.

The exercises serve as guidance; they will not be collected and will not count towards the course grade. There will not be any questions on quizzes or exams regarding the material contained in these notes per se. The content will of course be used in the process of solving differential equations, which will be on exams, quizzes and homework.

The first three sections are the essential part of the notes; we discuss complex numbers and their properties, the complex exponential function and the Fundamental Theorem of Algebra. We also have two appendices, for the more curious reader. Appendix A is about the roots of unity. Appendix B will be covered in the last part of the course when we study systems of differential equations.

1. COMPLEX NUMBERS

We start by considering the equation

$$(1.1) \quad x^2 + 1 = 0.$$

If we try to solve it, we run into the following:

$$x^2 = -1 \text{ and so } x = \sqrt{-1} \text{ or } x = -\sqrt{-1},$$

of course $\sqrt{-1}$ is not a real number, and so we introduce it as a new number:

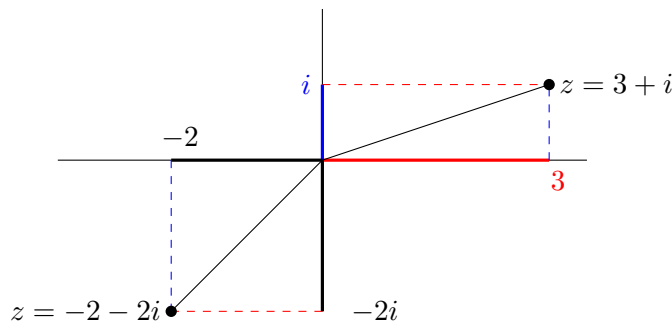


FIGURE 1. The geometric representations of $z = 3 + i$ and $z = -2 - 2i$.

$i = \sqrt{-1}$, which is called the imaginary number and is such that $i^2 = -1$.

Remark: Some engineering and physics books use $j = \sqrt{-1}$ instead of i .

Even though this number is called imaginary, we will see later, it does have great importance for solving differential equations which model real life phenomena.

A number of the form

$z = a + ib$, where a and b are real numbers, is called a complex number

a is called the real part of z , $a = \operatorname{Re} z$ and

b is called the imaginary part of z , $b = \operatorname{Im} z$.

The set of all complex numbers is denoted by $\mathbb{C}$. Examples of complex numbers:

$$z = 1 + 2i, \operatorname{Re} z = 1 \text{ and } \operatorname{Im} z = 2; \quad z = \frac{1}{2} - 8i, \operatorname{Re} z = \frac{1}{2} \text{ and } \operatorname{Im} z = -8;$$

$$z = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \operatorname{Re} z = \frac{1}{2} \text{ and } \operatorname{Im} z = \frac{\sqrt{3}}{2}; \quad z = \pi - e^2i, \operatorname{Re} z = \pi \text{ and } \operatorname{Im} z = -e^2.$$

While we associate real numbers to points on a line, we associate complex numbers to points on a plane, see Fig.1.

Exercises:

A.1 Find the real and imaginary parts of $z = 1 + 2i$ and $z = \pi + \sqrt{2}i$

A.2 Sketch the representation of $z = 2 + 5i$, $z = -1 + 7i$, $z = 1 - 5i$ on the plane

1.1. Operations with Complex Numbers. We can add, subtract, multiply and divide complex numbers (as long as the denominator is not equal to zero).

1.1.1. *Equality.* If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, we say that $z_1 = z_2$ if and only if $a_1 = a_2$ and $b_1 = b_2$. So

$$2 + 3i = 2 + 3i, \text{ but } 1 + 2i \neq 1 + 5i.$$

Also, we always write

$$z = 0 \text{ instead of } z = 0 + i0.$$

1.1.2. *Addition.* We add complex numbers as follows:

If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, their sum is defined to be

$$z_1 + z_2 = (a_1 + b_1) + i(a_2 + b_2).$$

For example:

$$\begin{aligned}(3 + 7i) + (2 + 5i) &= 5 + 12i. \\ (1 - \sqrt{3}i) + (2 + \sqrt{2}i) &= 3 + (\sqrt{2} - \sqrt{3})i.\end{aligned}$$

The sum of complex numbers resemble the sum of vectors; we just add the components. However, complex numbers are endowed with additional operations that vectors do not have:

1.1.3. *Multiplication.* We multiply complex numbers just like we multiply sums of real numbers, keeping in mind that $i^2 = -1$.

$$\begin{aligned}\text{if } z_1 &= a_1 + ib_1 \text{ and } z_2 = a_2 + ib_2, \text{ then} \\ z_1 z_2 &= (a_1 + ib_1)(a_2 + ib_2) = a_1(a_2 + ib_2) + ib_1(a_2 + ib_2) = \\ a_1 a_2 + ia_1 b_2 + ib_1 a_2 + i^2 b_1 b_2 &= a_1 a_2 - i^2 b_1 b_2 + i(a_1 b_2 + b_1 a_2) = (\text{because } i^2 = -1) \\ (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + a_2 b_1)\end{aligned}$$

For example

$$\begin{aligned}(2 + 3i)(1 + i) &= 2(1 + i) + 3i(1 + i) = 2 + 2i + 3i + 3i^2 = 2 - 3 + (2 + 3)i = -1 + 5i, \\ (1 + \sqrt{3}i)(1 - \sqrt{3}i) &= 1(1 - \sqrt{3}i) + i\sqrt{3}(1 - \sqrt{3}i) = 1 - \sqrt{3}i + \sqrt{3}i - (\sqrt{3})^2 i^2 = 4.\end{aligned}$$

1.1.4. *Properties of the operations.* It can be easily verified that complex addition and multiplication satisfy the usual commutation and distribution properties:

$$z_1 + z_2 = z_2 + z_1, \quad z_1 z_2 = z_2 z_1, \quad z(u + w) = zu + zw, \text{ and so on.}$$

1.1.5. *The Absolute Value.* If $z = a + ib$ we say that its absolute value, or modulus $|z|$, is equal to the length of the segment joining $(0, 0)$ and (a, b) . That is

$$\text{if } z = a + ib, \quad |z| = \sqrt{a^2 + b^2}.$$

For example,

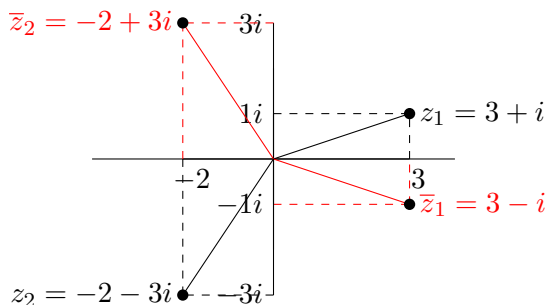
$$\begin{aligned}z &= 3 + 5i, \quad |z| = \sqrt{9 + 25} = \sqrt{34}, \\ z &= \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad |z| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1.\end{aligned}$$

1.1.6. *The Conjugate.* If $z = a + ib$, we define its conjugate $\bar{z} = a - ib$. That is, we just switch the sign of the imaginary part of z , see Fig. 2. For example,

$$\begin{aligned}\text{if } z &= 10 + 5i, \quad \bar{z} = 10 - 5i, \quad \text{if } z = 1 + \sqrt{3}i, \quad \bar{z} = 1 - \sqrt{3}i, \\ \text{if } z &= \sqrt{2} - 3i, \quad \bar{z} = \sqrt{2} + 3i, \quad \text{if } z = 3 - i\pi, \quad \bar{z} = 3 + i\pi.\end{aligned}$$

The product of z and $\bar{z}$ has a special property:

$$z = a + ib \quad \bar{z} = a - ib, \quad z\bar{z} = (a + ib)(a - ib) = a^2 - iab + iab - b^2 i^2 = a^2 + b^2 = |z|^2$$

FIGURE 2. The conjugates of $z_1 = 3 + i$ and $z_2 = -2 - 3i$

1.1.7. *Division.* We can divide a complex numbers, $z_1 = a_1 + ib_1$ by $z_2 = a_2 + ib_2$, provided $z_2 \neq 0$. We do the following:

$$\frac{z_1}{z_2} = \frac{a_1 + ib_1}{a_2 + ib_2} \text{ and we multiply and divide the fraction by } \overline{z_2} \text{ and we obtain}$$

$$\frac{z_1}{z_2} = \left(\frac{a_1 + ib_1}{a_2 + ib_2} \right) \left(\frac{a_2 - ib_2}{a_2 - ib_2} \right) = \frac{(a_1 + ib_1)(a_2 - ib_2)}{(a_2 + ib_2)(a_2 - ib_2)} = \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + i \frac{a_2 b_1 - a_1 b_2}{a_2^2 + b_2^2}.$$

Examples:

$$\frac{1 + 3i}{2 + 5i} = \left(\frac{1 + 3i}{2 + 5i} \right) \left(\frac{2 - 5i}{2 - 5i} \right) = \frac{(1 + 3i)(2 - 5i)}{(2 + 5i)(2 - 5i)} = \frac{17 + i}{29} = \frac{17}{29} + \frac{1}{29}i,$$

$$\frac{2 + i}{8 - 2i} = \left(\frac{2 + i}{8 - 2i} \right) \left(\frac{8 + 2i}{8 + 2i} \right) = \frac{(2 + i)(8 + 2i)}{(8 - 2i)(8 + 2i)} = \frac{14 + 12i}{68} = \frac{14}{68} + \frac{12}{68}i,$$

Exercises:

- B.1 Let $z_1 = 3 + 4i$ and $z_2 = 8 + 7i$. Compute $z_1 + z_2$, $z_1 z_2$, $\frac{z_2}{z_1}$ and $\frac{z_1}{z_2}$.
 B.2 Find the absolute value of $w_1 = 3 + 2i$ and $w_2 = 4 + 3i$.
 B.3 Describe the set of all complex numbers z such that $|z| = 3$.

2. EXPONENTIALS OF COMPLEX NUMBERS

We want to define the exponential function of a complex number. Recall that for a real number x ,

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n.$$

This series converges (absolutely) for all x . So it is natural to define

$$e^{ix} = \sum_{n=0}^{\infty} \frac{1}{n!} (ix)^n = \sum_{n=0}^{\infty} \frac{i^n}{n!} x^n.$$

We split the sum of this series into two parts (we can do this because it converges absolutely): when n is even ($n = 2k$) and when n is odd, ($n = 2k + 1$) and so

$$e^{ix} = \sum_{k=0}^{\infty} \frac{i^{2k}}{(2k)!} x^{2k} + \sum_{k=0}^{\infty} \frac{i^{2k+1}}{(2k+1)!} x^{2k+1}.$$

But notice that

$$\begin{aligned} i^{2k} &= (i^2)^k = (-1)^k \text{ and} \\ i^{2k+1} &= i(i^{2k}) = i(i^2)^k = (-1)^k i \end{aligned}$$

and so

$$e^{ix} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k)!} x^{2k} + i \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} x^{2k+1}.$$

If you check your Calculus book, you will see that

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k)!} x^{2k} \text{ and } \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} x^{2k+1},$$

and we deduce that

$$\boxed{e^{ix} = \cos x + i \sin x}, \text{ which is known as Euler's formula.}$$

Since $\sin(-x) = -\sin x$ and $\cos(-x) = \cos(x)$, we conclude that

$$\boxed{e^{-ix} = \cos x - i \sin x}$$

and we obtain the following useful formulas

$$\boxed{\cos x = \frac{1}{2}(e^{ix} + e^{-ix}) \text{ and } \sin x = \frac{1}{2i}(e^{ix} - e^{-ix}).}$$

It may be somewhat surprising that the exponential function and the sine and cosine functions could be related. We usually associate exponential functions with rapid growth or decay, and associate sine and cosine with oscillations. But this is one of the key reasons complex numbers are important and so useful for solving differential equations. In general, complex exponentials can be associated with functions which both oscillate and exponentially grow or decay, see for example Fig.8 and Fig.9. We will discuss these later in Appendix B.

The Euler's formula for $x = \pi$ gives

$$\boxed{e^{i\pi} + 1 = 0}, \text{ which is called Euler's identity.}$$

This formula is somewhat mysterious because it relates the numbers i , e , π one and zero: i is the imaginary number we just defined, π comes from geometry, and is defined to be the ratio of the area of a circle and the square of its radius, e is Euler's number which is defined in terms of a limit $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$. Some consider this to be the most beautiful mathematical formula because it connects these seemingly unrelated numbers, and the numbers one and zero, in a single equation.

However, we still need to verify that e^{ix} behaves as an exponential function, namely that

$$e^{ix} e^{iy} = e^{i(x+y)}.$$

So we start from the left hand side and use the multiplication rule for complex numbers, and we obtain

$$\begin{aligned} e^{ix} e^{iy} &= (\cos x + i \sin x)(\cos y + i \sin y) = \\ &= (\cos x \cos y - \sin x \sin y) + i(\cos x \sin y + \sin x \cos y) \end{aligned}$$

But we recall the following trigonometric identities:

$$(2.1) \quad \begin{aligned} \cos(x+y) &= \cos x \cos y - \sin x \sin y, \text{ and} \\ \sin(x+y) &= \sin x \cos y + \sin y \cos x, \end{aligned}$$

and so we conclude that

$$e^{ix} e^{iy} = \cos(x+y) + i \sin(x+y) = e^{i(x+y)}.$$

So, it in fact behaves like an exponential. In particular,

$$(e^{ix})^m = (\cos x + i \sin x)^m = e^{imx} = \cos(mx) + i \sin(mx),$$

which is known as de Moivre's formula.

In general, if $z = a + ib$ we define

$$e^z = e^{a+ib} = e^a e^{ib} = e^a (\cos b + i \sin b),$$

and it is the case that

$$e^{z_1} e^{z_2} = e^{z_1+z_2} \text{ and in particular } e^z e^{-z} = e^0 = 1, \text{ so } e^{-z} = \frac{1}{e^z}.$$

For example

$$\begin{aligned} e^{2+3i} &= e^2 (\cos 3 + i \sin 3), \quad e^{5+\pi i} = e^5 (\cos \pi + i \sin \pi) = -e^5, \\ (e^{3+5i})(e^{6-8i}) &= e^{(3+5i)+(6-8i)} = e^{9-3i}, \\ \frac{1}{e^{1+\pi i}} &= e^{-(1+i\pi)} = e^{-1} e^{-\pi i} = e^{-1} (\cos \pi - i \sin \pi) = -e^{-1}. \end{aligned}$$

Exercises:

- C.1 Find real numbers a and b such that $e^{2+i\pi} = a + bi$.
 C.2 Find real numbers a and b such that $e^{\ln 3 + i\frac{\pi}{2}} = a + bi$.
 C.3 Compute the product $e^{1+2i} e^{5+7i}$.
 C.4 Compute the quotient $\frac{e^{3+4i}}{e^{1+8i}}$.

2.1. Exponential functions and their derivatives. We learned in Calculus that if a is a real number,

$$\frac{d}{dx} e^{ax} = a e^{ax} \text{ and in general } \frac{d^n}{dx^n} e^{ax} = a^n e^{ax}, \quad n = 1, 2, \dots$$

If $za + ib$ is a complex number, we define the associated exponential function

$$f(x) = e^{zx} = e^{(a+ib)x} = e^{ax+ibx} = e^{ax} (\cos(bx) + i \sin(bx)), \quad x \text{ is a real number.}$$

We can differentiate $f(x)$ using the product rule

$$\begin{aligned} f'(x) &= \frac{df}{dx}(x) = \frac{d}{dx} (e^{ax} (\cos(bx) + i \sin(bx))) + e^{ax} \frac{d}{dx} (\cos(bx) + i \sin(bx)) = \\ &= a e^{ax} (\cos(bx) + i \sin(bx)) + b e^{ax} (-\sin(bx) + i \cos(bx)) = \\ &= e^{ax} (a \cos(bx) - b \sin(bx)) + i e^{ax} (a \sin(bx) + b \cos(bx)) = \\ &= e^{ax} (a + ib) (\cos(bx) + i \sin(bx)) = z e^{zx}. \end{aligned}$$

So we conclude that, just like in the case where z is a real number,

$$\boxed{\frac{d}{dx} e^{zx} = z e^{zx}} \quad z \text{ complex and } x \text{ real.}$$

and of course in general, for all derivatives

$$\boxed{\frac{d^n}{dx^n} e^{zx} = z^n e^{zx}, \quad n = 1, 2, \dots} \quad z \text{ complex and } x \text{ real.}$$

As we will see later, this is quite useful for solving differential equations.

Exercises:

D.1 Let $f(x) = e^{(3+2i)x}$. Find $f'(x)$ and $f''(x)$

D.2 Let $g(x) = e^{(1+i)x}$. Find $g^{(3)}(x)$ (the third derivative of $g(x)$.)

2.2. More About Conjugates. The operation of taking conjugates behaves well under additions, products, quotients and exponentials. Namely

$$(2.2) \quad \begin{aligned} \overline{z_1 + z_2} &= \overline{z_1} + \overline{z_2}, \\ \overline{z_1 z_2} &= \overline{z_1} \overline{z_2}, \\ \overline{\left(\frac{z_1}{z_2}\right)} &= \frac{\overline{z_1}}{\overline{z_2}}, \text{ provided } z_2 \neq 0, \\ \overline{e^z} &= e^{\overline{z}}. \end{aligned}$$

We will verify the last one, and the reader should verify the other properties. One can see that if $z = a + ib$,

$$e^z = e^a(\cos b + i \sin b),$$

and so

$$\overline{e^z} = e^a(\cos b - i \sin b),$$

but since $\cos(-b) = \cos b$ and $\sin(-b) = -\sin b$, we have

$$\overline{e^z} = e^a(\cos(-b) + i \sin(-b)) = e^a \cdot e^{-ib} = e^{a-ib} = e^{\overline{z}}.$$

Exercises:

E.1 Find the real and the imaginary parts of $e^{(3+4i)x}$.

E.2 Find the real and the imaginary parts of $e^{(2-3i)x}$.

E.3 Verify that the functions $h_1(x) = e^{(2+i)x}$ and $h_2(x) = e^{(2-i)x}$ satisfy the differential equation $h''(x) - 4h'(x) + 5h(x) = 0$

E.4 Compute the real and imaginary parts of $h_1(x)$ and $h_2(x)$.

E.5 Use the previous exercise to verify that $e^{2x} \cos(x)$ and $e^{2x} \sin(x)$ also satisfy the differential equation $h''(x) - 4h'(x) + 5h(x) = 0$.

2.3. The polar representation. Complex numbers are represented by points on the plane and so one can use polar coordinates to represent them. If $z = a + ib \neq 0$, then

$$|z| = \sqrt{a^2 + b^2} > 0,$$

and if we let $r = |z|$, we can write

$$z = r \left(\frac{a}{r} + i \frac{b}{r} \right), \text{ there exists an angle } \theta \text{ in } [0, 2\pi) \text{ such that}$$

$$\frac{a}{r} = \frac{a}{\sqrt{a^2 + b^2}} = \cos \theta \text{ and } \frac{b}{r} = \frac{b}{\sqrt{a^2 + b^2}} = \sin \theta, \text{ see Fig. 3,}$$

and so by Euler's formula we can write

$$z = |z|(\cos \theta + i \sin \theta) = re^{i\theta}.$$

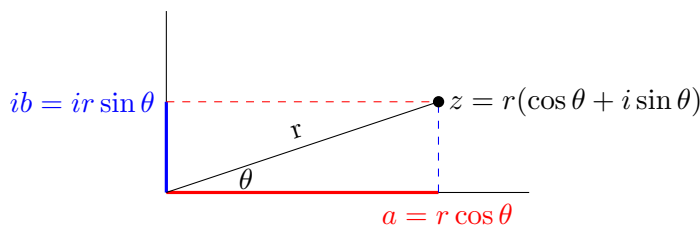


FIGURE 3. The polar representation of $z = a + ib$

Examples:

$$z = 2 + 3i, \quad r = |z| = \sqrt{4 + 9} = \sqrt{13}, \quad \cos \theta = \frac{2}{\sqrt{13}}, \quad \sin \theta = \frac{3}{\sqrt{13}},$$

Since $\cos \theta$ and $\sin \theta$ are both positive, θ is in the first quadrant and if one uses a calculator one finds the approximate value of $\theta \approx 0.2328$, and we write

$$z = \sqrt{13} e^{i\theta}, \quad \text{with } \cos \theta = \frac{2}{\sqrt{13}}, \quad \sin \theta = \frac{3}{\sqrt{13}}; \quad \theta \approx 0.2328.$$

Consider the following

$$z = 3 - 3\sqrt{3}i, \quad |z| = r = \sqrt{9 + 27} = \sqrt{36} = 6,$$

$$\cos \theta = \frac{3}{6} = \frac{1}{2}, \quad \text{and } \sin \theta = -\frac{3\sqrt{3}}{6} = -\frac{\sqrt{3}}{2}.$$

Notice that $\cos \theta$ is positive and $\sin \theta$ is negative, so θ is in the fourth quadrant, in fact

$$\theta = \frac{5\pi}{3} \quad \text{and so } z = 6 e^{i\frac{5\pi}{3}}.$$

Similarly,

$$z = -3 + 3i, \quad |z| = r = \sqrt{9 + 9} = 3\sqrt{2},$$

$$\cos \theta = -\frac{3}{3\sqrt{2}} = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}, \quad \sin \theta = \frac{3}{3\sqrt{2}} = \frac{\sqrt{2}}{2},$$

and so $\cos \theta < 0$ and $\sin \theta > 0$, so θ is in the second quadrant, $\pi/2 < \theta < \pi$, and we obtain $\theta = \frac{3\pi}{4}$. Therefore

$$z = 3\sqrt{2} e^{i\frac{3\pi}{4}}.$$

The polar representation formula makes it possible to find powers of a complex number:

$$(2.3) \quad \begin{aligned} &\text{if } z = re^{i\theta}, \quad z \neq 0, \quad \text{and } m \text{ is a real number,} \\ &\text{then } z^m = r^m e^{im\theta} = r^m (\cos(m\theta) + i \sin(m\theta)). \end{aligned}$$

For example, let's say we want to compute $(1 + i\sqrt{3})^8$ and $(1 + i\sqrt{3})^{\frac{1}{2}}$. Of course, in the first case we can multiply it eight times, but it is much simpler if we use its polar representation for both cases. We first find the polar representation of $z = 1 + i\sqrt{3}$,

$$|z| = \sqrt{1 + 3} = 2, \quad \cos \theta = \frac{1}{2} \quad \text{and } \sin \theta = \frac{\sqrt{3}}{2}, \quad \text{so } \theta = \frac{\pi}{3} \quad \text{and } z = 2e^{i\frac{\pi}{3}}.$$

To find z^8 we compute

$$(1 + \sqrt{3})^8 = (2e^{i\frac{\pi}{3}})^8 = 2^8 e^{i\frac{8\pi}{3}}$$

But $\frac{8\pi}{3} = 2\pi + \frac{2\pi}{3}$, and so $e^{\frac{8\pi i}{3}} = e^{\frac{2\pi i}{3}}$. Since $\frac{2\pi}{3}$ is in the second quadrant,

$$\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} \text{ and } \sin\left(\frac{4\pi}{3}\right) = \frac{\sqrt{3}}{2},$$

and so

$$(1 + \sqrt{3}i)^8 = 2^8 \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2^7(-1 + i\sqrt{3}) = 128(-1 + i\sqrt{3}).$$

On the other hand,

$$(1 + i\sqrt{3})^{\frac{1}{2}} = (2e^{i\frac{\pi}{3}})^{\frac{1}{2}} = \sqrt{2}e^{i\frac{\pi}{6}} = \sqrt{2}\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right) = \sqrt{2}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right).$$

Exercises:

F.1 Find the polar representation of $z = 1 + i$ and use it to compute $(1 + i)^{10}$ and $(1 + i)^{\frac{1}{5}}$

3. THE FACTORIZATION OF POLYNOMIALS

In the process of solving certain differential equations we will need to factor polynomials, which means we want to write them as a product of powers of polynomials of degree one, such as

$$p(x) = a(x - x_1)^{m_1}(x - x_2)^{m_2} \dots (x - x_k)^{m_k}, \quad x_a \neq x_b, \quad a, b = 1, 2, \dots, k.$$

We say that x_k is a root of a polynomial $p(x)$ if $p(x_k) = 0$. In the case above, the roots of $p(x)$ are $x_1, x_2, \dots, x_k$. The numbers $m_1, m_2, \dots, m_k$ are called the multiplicity of the corresponding root.

For example, in the case of

$$p(x) = (x - 1)^3(x + 2)^2(x - 3)^4$$

the roots are $x = 1$ and $x = -2$ and $x = 3$, and 3, 2 and 4 are their respective multiplicities.

We will restrict ourselves to discussing only polynomials with real coefficients. However, as we have seen in the case of $p(x) = x^2 + 1$, even polynomials with real coefficients may have complex roots.

3.1. Polynomials of degree two. Polynomials of degree two are of the form

$$p(x) = ax^2 + bx + c,$$

and they are also called quadratic polynomials.

Certain polynomials of degree two are easily factored, for example:

$$x^2 + 5x + 6 = (x + 2)(x + 3).$$

In this case, the roots of the polynomial are -2 and -3 , and both have multiplicity one. Roots of multiplicity one are often called simple roots.

The polynomial

$$p(x) = x^2 - 2x + 1 = (x - 1)^2.$$

has only one root, 1, but it has multiplicity two, and we call it a double root.

Sometimes it is not so easy to factor quadratic polynomials, but fortunately we have a formula to find the roots of quadratic polynomials:

(3.1) If $a \neq 0$ the two roots of the equation $ax^2 + bx + c = 0$ are

$$z_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } z_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a},$$

and we can write the polynomial

$$p(x) = ax^2 + bx + c = a(x - z_1)(x - z_2).$$

We are using z_1 and z_2 instead of x_1 and x_2 to denote the roots of $p(x)$ to emphasize that these may be complex numbers. As we have already seen, quadratic polynomials, even if their coefficients are real, may have complex roots. For example, we have

$$x^2 + 1 = (x - i)(x + i). \text{ The roots are } z_1 = i, \quad z_2 = -i.$$

In general, a polynomial $p(x) = ax^2 + bx + c$ has complex roots, if the term inside the square root in formula (3.1) is negative, that is if

$$D = b^2 - 4ac < 0, \quad \text{where } D \text{ is called the discriminant of the polynomial.}$$

Example: If $p(x) = x^2 + 1$, $a = 1$, $b = 0$ and $c = 1$, so $D = b^2 - 4ac = -4$. In fact, as we have seen above, its roots are i and $-i$.

Example: If $p(x) = x^2 + 4x + 5$, $a = 1$, $b = 4$ and $c = 5$, so $D = b^2 - 4ac = -4$. We can find its roots using formula (3.1):

$$\begin{aligned} z_1 &= \frac{-4 + \sqrt{16 - 20}}{2} = \frac{-4 + \sqrt{-4}}{2} = \frac{-4 + \sqrt{4(-1)}}{2} = \frac{-4 + 2\sqrt{-1}}{2} = -2 + i, \\ z_2 &= \frac{-4 - \sqrt{16 - 20}}{2} = \frac{-4 - \sqrt{-4}}{2} = \frac{-4 - \sqrt{4(-1)}}{2} = \frac{-4 - 2\sqrt{-1}}{2} = -2 - i, \end{aligned}$$

Notice that $z_2 = \bar{z}_1$. So we conclude that

$$\begin{aligned} p(x) &= x^2 + 4x + 5 = (x - z_1)(x - z_2) = \\ &= (x - (-2 + i))(x - (-2 - i)) = (x + 2 - i)(x + 2 + i). \end{aligned}$$

So we come to the following conclusion: A quadratic polynomial $p(x)$ with real coefficients can always be factored

$$p(x) = ax^2 + bx + c = a(x - z_1)(x - z_2),$$

with z_1 and z_2 given by equation (3.1). The nature of the roots of $p(x)$ is determined by the discriminant $D = b^2 - 4ac$.

1. If $D > 0$, z_1 and z_2 are real and distinct,
2. If $D = 0$, there is only one real root $z_1 = z_2 = -\frac{b}{2a}$, but with multiplicity two,
3. If $D < 0$, the polynomial has two distinct complex roots z_1 and z_2 . Moreover, $z_2 = \bar{z}_1$.

So, for polynomials of degree two with real coefficients, the complex roots come in pairs, in the sense that if z is a root, $\bar{z}$ is also a root. This is part of a general pattern:

Important Fact: Complex roots of polynomials $p(x)$ with real coefficients come in pairs in the sense that, if z is a root of the polynomial, so is $\bar{z}$. In other words, if $p(z) = 0$, then $p(\bar{z}) = 0$. One

can see that it is the case because if

$$p(x) = a_n x^n + a_{n-1} x^{n-1} \dots a_1 x + a_0, \text{ and}$$

$$p(z) = a_n z^n + a_{n-1} z^{n-1} \dots a_1 z + a_0 = 0,$$

and the coefficients $a_n, a_{n-1}, \dots, a_0$ are real numbers. In this case $\overline{a_j} = a_j$, and then because of (2.2)

$$\overline{a_j z^j} = \overline{a_j} \overline{z^j} = a_j \overline{z}^j$$

and therefore

$$\overline{p(z)} = a_n \overline{z}^n + a_{n-1} \overline{z}^{n-1} \dots a_1 \overline{z} + a_0 = p(\overline{z}) = 0.$$

3.2. Polynomials of degree three. It is not as easy to factor polynomials of degree three and higher. There are formulas for solving polynomial equations of degree three and four, but not for degree 5 and higher.

The remark above tells us that if $p(x)$ has degree three, *and its coefficients are real*, it may not have three complex roots, or only one complex root, because complex roots always occur in pairs z and $\overline{z}$. So it either has three real roots (counted with multiplicity), or one real and two complex roots.

Example: We want to factor the polynomial $p(x) = x^3 - 7x^2 + 14x - 8$, and we are told that $x = 1$ is one of its roots. So we must have

$$p(x) = x^3 - 7x^2 + 14x - 8 = (x - 1)(x^2 + ax + b).$$

We can use the Euclidean algorithm to divide the polynomials $\frac{x^3 - 7x^2 + 14x - 8}{x - 1}$, just as we divide integers, but one can also perform the multiplication, equate the coefficients and find a and b . For example, in this case

$$x^3 - 7x^2 + 14x - 8 = (x - 1)(x^2 + ax + b) =$$

$$x(x^2 + ax + b) - (x^2 + ax + b) = x^3 + (a - 1)x^2 + (b - a)x - b.$$

We equate the coefficients of the first and the last equation and we find that

$$a - 1 = -7, \text{ so } a = -6, \quad b - a = 14, \text{ and } b = 8$$

and so we have

$$p(x) = x^3 - 7x^2 + 14x - 8 = (x - 1)(x^2 - 6x + 8) = (x - 1)(x - 2)(x - 4).$$

Example: Factor $p(x) = x^3 - 3x + 2$, given that $x = 1$ is a root. If we follow the same method, we find that

$$p(x) = x^3 - 3x + 2 = (x - 1)(x^2 + x - 2) = (x - 1)(x - 1)(x + 2) = (x - 1)^2(x + 2).$$

In this case we see that the root $x = 1$ has multiplicity two. The polynomial has three roots: 1, 1, -2; we count the root 1 twice.

Example: We can also have roots of multiplicity three:

$$p(x) = x^3 - 3x^2 + 3x - 1 = (x - 1)^3,$$

and $x = 1$ is a root of multiplicity three.

This gets harder when we have complex roots. For example, factor the polynomial $p(x) = x^3 - 8$. We notice that it has one root $x = 2$. Again, we follow the same method used above and we find

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4).$$

The roots of $x^2 + 2x + 4$ are

$$z_1 = \frac{-2 + \sqrt{-12}}{2} = \frac{-2 + 2\sqrt{3}i}{2} = -1 + \sqrt{3}i,$$

$$z_2 = \frac{-2 - \sqrt{-12}}{2} = \frac{-2 - 2\sqrt{3}i}{2} = -1 - \sqrt{3}i.$$

As we had discussed before, $z_2 = \bar{z}_1$. So

$$x^3 - 8 = (x - 2)(x - (-1 + \sqrt{3}i))(x - (-1 - \sqrt{3}i)).$$

In this case, the polynomial has one real root and two complex ones.

3.3. Polynomials of degree four. As one can expect, it is harder to factor polynomials of degree four, but some cases are not so difficult and we can reduce them to products of polynomials of degree two. For example

$$x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x - 2i)(x + 2i).$$

Polynomials of degree which have no terms in x or x^3 , such as

$$p(x) = ax^4 + bx^2 + c.$$

are called biquadratic. We can find its roots by first reducing it to a quadratic polynomial by setting $y = x^2$, and we have

$$q(y) = ay^2 + by + c = a(y - w_1)(y - w_2)$$

Since $x^2 = y$, we have $x^2 = w_1$ and $x^2 = w_2$. When w_1 and w_2 are real, we have four roots of the original polynomial $p(x)$,

$$z_1 = \sqrt{w_1}, \quad z_2 = -\sqrt{w_1}, \quad z_3 = \sqrt{w_2}, \quad z_4 = -\sqrt{w_2}.$$

When w_1 and (or) w_2 are complex, one can use equation (2.3) with $m = \frac{1}{2}$ to compute $\sqrt{w_j}$, $j = 1, 2$. This will be discussed in more details in Appendix A.

Example: Factor $p(x) = x^4 - 5x^2 - 36$. First we set $y = x^2$, and we obtain

$$q(y) = y^2 - 5y - 36 = (y - 9)(y + 4).$$

Since $x^2 = y$, we have $x^2 = 9$ and $x^2 = -9$, and so

$$z_1 = \sqrt{9} = 3, \quad z_2 = -\sqrt{9} = -3,$$

$$z_3 = \sqrt{-4} = 2i, \quad z_4 = -\sqrt{-4} = -2i.$$

So we have

$$p(x) = x^4 - 5x^2 - 36 = (x - 3)(x + 3)(x + 2i)(x - 2i).$$

3.4. Polynomials of higher degree. In general it is very difficult to factor polynomials of degree five and higher, but we will also encounter polynomials which have already been partially factored and we just need to find their roots and the corresponding multiplicity.

Example: Consider the polynomial of degree 21:

$$p(x) = (x - 1)^2(x - 2)^5(x^2 + 3x + 3)^3(x^2 + 4x + 5)^4.$$

We can expand this further by finding the complex roots of the second degree polynomials. The roots of $x^2 + 4x + 5$ are given above, and we can find the roots of $x^2 + 3x + 3$ by using the formula above, and they are

$$z_1 = \frac{-3 + i\sqrt{3}}{2} \text{ and } z_2 = \frac{-3 - i\sqrt{3}}{2}.$$

Again, notice that $z_2 = \bar{z}_1$. Therefore one may rewrite

$$p(x) = (x-1)^2(x-2)^5\left(x - \frac{-3+i\sqrt{3}}{2}\right)^3\left(x - \frac{-3-i\sqrt{3}}{2}\right)^3(x+2-i)^4(x+2+i)^4.$$

The multiplicity of the roots are

$$\begin{aligned} z_1 = 1 & \text{ has multiplicity } 2, \quad z_2 = 2 \text{ has multiplicity } 5, \\ z_3 = \frac{-3+i\sqrt{3}}{2} & \text{ and } z_4 = \frac{-3-i\sqrt{3}}{2} \text{ have multiplicity } 3, \\ z_5 = -2+i & \text{ and } z_6 = -2-i \text{ have multiplicity } 4. \end{aligned}$$

If we add the multiplicities we obtain $2 + 5 + 3 + 3 + 4 + 4 = 21$, which is equal to the degree of the polynomial.

Example: Consider the polynomial of degree 28 partially factored in the following way

$$p(x) = (x-1)^3(x-7)^4(x-8)^5(x^2+4x+5)^3(x^2+9)^5.$$

Its real roots are

$$z_1 = 1 \text{ with multiplicity } 3, \quad z_2 = 7 \text{ with multiplicity } 4, \text{ and } z_3 = 8 \text{ with multiplicity } 5,$$

and its complex roots are the roots of the quadratic polynomials

$$\begin{aligned} z_4 = -2+2i \text{ and } \bar{z}_5 = -2-2i & \text{ each with multiplicity } 3, \text{ which correspond to } x^2+4x+5, \\ z_6 = 3i \text{ and } \bar{z}_7 = -3i & \text{ each with multiplicity } 5, \text{ which correspond to } x^2+9. \end{aligned}$$

The sum of the multiplicity of the roots is equal to 28, which is the degree of the polynomial.

According to the examples we have seen, polynomials of degree 2 have two roots, which may be complex, provided we count them with multiplicity. Polynomials of degree three have three roots, again, complex and counted with multiplicity, and so on. Moreover, if the polynomial has only real coefficients the roots appear in pairs: if z is a root, so is $\bar{z}$.

One wonders if that is the case in general, and if a polynomial of degree n has n complex roots, provided they are counted with multiplicity. The answer is yes, and it is given by the following result:

The Fundamental Theorem of Algebra: A polynomial of degree n ,

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

with complex or real coefficients $a_n, a_{n-1}, \dots, a_0$, ($a_n \neq 0$, otherwise its degree would not be equal to n) can be factored in one unique way as

$$p(x) = a_n(x-z_1)^{m_1} \dots (x-z_k)^{m_k}$$

where $z_1, \dots, z_k$ are the distinct roots of $p(x)$, and m_1 is the multiplicity of the root z_1 , m_2 is the multiplicity of z_2 and so on, and m_k is the multiplicity of the root z_k , and the sum

$$m_1 + m_2 + \dots + m_k = n \quad (\text{the degree of the polynomial}).$$

As mentioned above, we encounter polynomials with real coefficients $a_n, a_{n-1}, \dots, a_0$, which are factored in the following way:

$$p(x) = a_n(x-x_1)^{m_1} \dots (x-x_k)^{m_k}(x^2+b_1x+c_1)^{p_1}(x^2+b_2x+c_2)^{p_2} \dots (x^2+b_jx+c_j)^{p_j}.$$

Here $x_1, \dots, x_k$ are the **real** roots of $p(x)$, and m_1 is the multiplicity of the root x_1 , m_2 is the multiplicity of x_2 and so on, and m_k is the multiplicity of the root x_k . The quadratic polynomials

$$q_j(x) = x^2 + b_jx + c_j,$$

do not have real roots, and

$$x^2 + b_i x + c_i = (x - z_i)(x - \bar{z}_i), \quad z_i \text{ complex},$$

The multiplicity of the complex roots z_i and $\bar{z}_i$ are given by p_i . The sum of the multiplicities of the real roots and the complex roots is equal to n , which is the degree of the polynomial,

$$m_1 + m_2 + \dots + m_k + 2p_1 + 2p_2 + \dots + 2p_j = n,$$

the p_j count twice because they correspond to roots z_j and $\bar{z}_j$.

Exercises:

G.1 Find the roots of the polynomial $(x^2 + 3x + 2)^3(x^2 - 4)^2$ and their respective multiplicities.

G.2 Find the roots of the polynomial $(x^3 + x^2 - x - 1)^2$ and their respective multiplicities.

APPENDIX A. THE ROOTS OF UNITY

In this section we will find the roots of polynomial equations of the form

$$(A.1) \quad z^m - 1 = 0, \text{ where } m \text{ is a positive integer.}$$

As we have just learned, this equation has m roots counted with multiplicity and if z is a complex solution, so is $\bar{z}$. To solve this equation we will use the polar representation of z , and so we want to find all $z = re^{i\theta}$ which satisfies equation (A.1). Therefore we must have

$$(A.2) \quad z^m = r^m e^{im\theta} = 1 = e^{i0}.$$

We first conclude that

$$r = 1.$$

However, we have to be more careful when we try to find the angle θ . One solution of course is $\theta = 0$, but this is not the only one which satisfies (A.2). The functions $\sin \theta$ and $\cos \theta$ are periodic of period 2π and so is $e^{i\theta}$. Hence

$$(A.3) \quad m\theta = 2k\pi,$$

would also satisfy the equation for all integers k . However, since $r^m e^{im\theta}$ is the polar representation of z^m , $m\theta$ has to be in the interval $[0, 2\pi)$, so we find m values for θ :

$$m\theta = 2k\pi, \quad k = 0, 1, \dots, m-1,$$

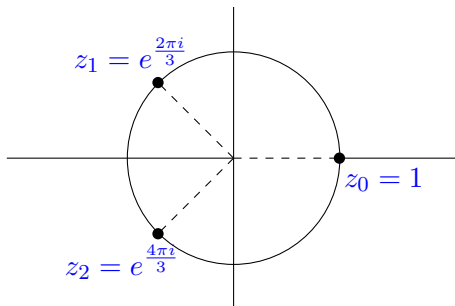
and any additional values of k would give one of these solutions plus an integer multiple of 2π . So we have found m solutions of (A.1):

$$(A.4) \quad z_0 = 1, \quad z_1 = e^{\frac{2\pi i}{m}}, \quad z_2 = e^{\frac{4\pi i}{m}}, \dots, \quad z_{m-1} = e^{\frac{2(m-1)\pi i}{m}}.$$

Notice that $|z_j| = 1$, for $j = 0, 1, \dots, m-1$, so these are points on the unit circle which divide it into m equal parts, see Fig.4 and Fig.5 for the cases $m = 3$ and $m = 4$.

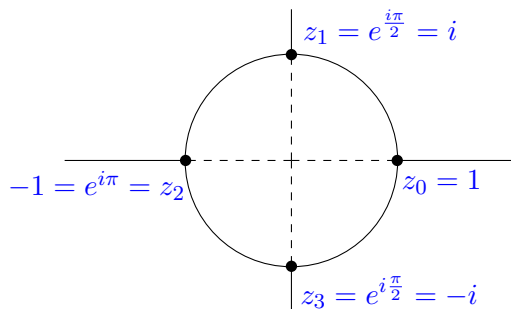
For example, the solutions of $z^3 = 1$ are, see Fig.4:

$$\begin{aligned} z_0 = 1, \quad z_1 &= e^{\frac{2\pi i}{3}} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \\ \text{and } z_2 &= e^{\frac{4\pi i}{3}} = \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i. \end{aligned}$$

FIGURE 4. The solutions of $z^3 = 1$.

The solutions of $z^4 = 1$ are, see Fig. 5:

$$\begin{aligned} z_0 &= 1, \quad z_1 = e^{\frac{2\pi i}{4}} = e^{\frac{\pi i}{2}} = i, \\ z_2 &= e^{\frac{4\pi i}{4}} = e^{\pi i} = -1, \quad z_3 = e^{\frac{6\pi i}{4}} = e^{\frac{3\pi i}{2}} = -i. \end{aligned}$$

FIGURE 5. The solutions of $z^4 = 1$.

Next we want to find all the solutions of equations of the form

$$(A.5) \quad z^m - w = 0, \quad w \neq 0.$$

Of course one can follow the same method used to solve the equation for $w = 1$, but one just needs to observe that if $u^m = 1$, then $z = w^{\frac{1}{m}}u$ satisfies $z^m = w$. So the m roots of (A.5) are given by $w^{\frac{1}{m}}$ times the roots of $z^m - 1 = 0$, namely:

$$(A.6) \quad z_0 = w^{\frac{1}{m}}, \quad z_1 = w^{\frac{1}{m}} e^{\frac{2\pi i}{m}}, \quad z_2 = w^{\frac{1}{m}} e^{\frac{4\pi i}{m}}, \dots, \quad z_{m-1} = w^{\frac{1}{m}} e^{\frac{2(m-1)\pi i}{m}}.$$

Recall that $w^{\frac{1}{m}} = \rho^{\frac{1}{m}} e^{i\frac{\alpha}{m}}$, where $w = \rho e^{i\alpha}$ is the polar representation of w .

For example, find all z such that

$$(A.7) \quad z^4 = i.$$

We first find the polar representation of $w = i = e^{\frac{\pi i}{2}}$. Then

$$w^{\frac{1}{4}} = e^{i\frac{\pi}{8}},$$

and according to (A.6), the four solutions of (A.7) are

$$z_0 = e^{\frac{\pi i}{8}}, \quad z_1 = e^{\frac{\pi i}{8}} e^{\frac{\pi i}{2}} = e^{\frac{5\pi i}{8}}, \quad z_2 = e^{\frac{\pi i}{8}} e^{\pi i} = e^{\frac{9\pi i}{8}}, \quad z_3 = e^{\frac{\pi i}{8}} e^{\frac{3\pi i}{2}} = e^{\frac{13\pi i}{8}}.$$

APPENDIX B. MORE ABOUT COMPLEX EXPONENTIAL FUNCTIONS

We give examples that show how complex numbers are important in the study of differential equation which describe physical phenomena. This is covered, with somewhat different notation, in the study of systems of differential equations.

We will encounter functions of the type $f(t) = e^{(a+ib)t}$ which are solutions of differential equations. As we have discussed above, they obey the same rules of differentiation, namely:

$$\frac{d^n}{dt^n} e^{zt} = z^n e^{zt}, \text{ for } z \text{ in } \mathbb{C} \text{ and } n = 1, 2, \dots$$

recall that $\mathbb{C}$ denotes the set of complex numbers. For example if $f(t) = e^{ibt}$, then

$$\begin{aligned} f'(t) &= \frac{d}{dt} e^{ibt} = ibe^{ibt}, \\ f''(t) &= ib(ibe^{it}) = -b^2 e^{ibt} = -b^2 f(t) \end{aligned}$$

and so $f(t) = e^{ibt}$ satisfies the following differential equation

$$f''(t) + b^2 f(t) = 0.$$

The function $g(t) = e^{-ibt}$ also satisfies the same differential equation.

For each t , $f(t)$ defines a point on the plane and as t varies, this describes a curve on the plane. We can sketch the curve defined by $f(t)$. Since,

$$f(t) = \cos(bt) + i \sin(bt),$$

the point on the plane associated with $f(t)$ has coordinates $x(t) = \cos(bt)$ and $y(t) = \sin(bt)$. If one thinks of t as time, and $b > 0$, this describes the counterclockwise periodic motion of a point on the unit circle with angular speed b . In the case when $b < 0$, it also describes a periodic motion around the circle, but with opposite orientation, see Fig.6.

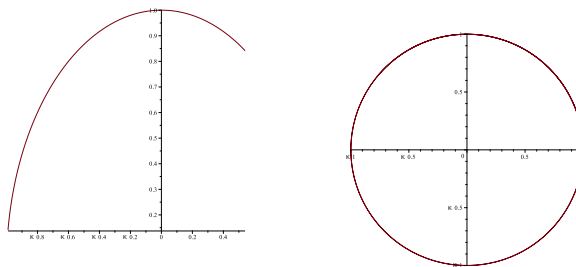


FIGURE 6. The curve on the plane defined by $f(t) = e^{3it}$ for $\frac{1}{3} \leq t \leq 1$ on the left and for $0 \leq t \leq 8$ on the right

Similarly, one can check that $h(t) = e^{(a+ib)t}$

$$h''(t) - 2ah'(t) + (a^2 + b^2)h(t) = 0,$$

and $F(t) = e^{(a-ib)t}$ also satisfies this differential equation

Exercise: Check that $f(t) = e^{(-2+i)t}$ satisfies $f''(t) + 4f'(t) + 5f(t) = 0$.

One can sketch the curve described by $F(t) = e^{(1+10i)t}$ as t varies.

$$F(t) = e^{(1+10i)t} = e^t(\cos(10t) + i \sin(10t)),$$

The coordinates of the point on the plane associated with $F(t)$ are

$$x(t) = e^t \cos(10t), \quad y(t) = e^t \sin(10t),$$

This curve is an outward spiral. While the term $e^{i10t} = \cos(10t) + i \sin(10t)$ describes the counterclockwise periodic motion of a point around the circle with angular speed equal to 10, the term e^t , makes its absolute value grow with time, see Fig.7. Similarly, one can sketch the graph of $F(t) = e^{(-1+10i)t}$.

$$F(t) = e^{(-1+10i)t} = e^{-t}(\cos(10t) + i \sin(10t)),$$

Now the term e^{-t} , makes its absolute value go to zero as t gets large and positive, see Fig.7.

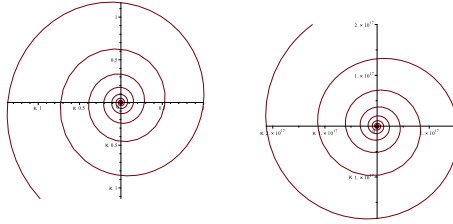


FIGURE 7. The graph of $e^{(1+10i)t}$ on the left and the one of $e^{(-1+10i)t}$ on the right for $-40 \leq t \leq 40$.

The real and imaginary parts of the complex exponentials also satisfy the same differential equation. For example, according to what we have said above

$$f(t) = e^{(-1+10i)t} = e^{-t} \cos(10t) + i e^{-t} \sin(10t)$$

satisfies the equation

$$f'' + 2f' + 101f = 0.$$

But it is also true that the real and imaginary parts of $f(t)$ also satisfy the same equation, and we now get two real solutions

$$y_1(t) = e^{-t} \cos(10t) \text{ and } y_2(t) = e^{-t} \sin(10t).$$

The graphs in Fig.8 appear in the study of motion with damping. When the real part is positive, the amplitude of the oscillations get larger with time, and one can see in the following graph:

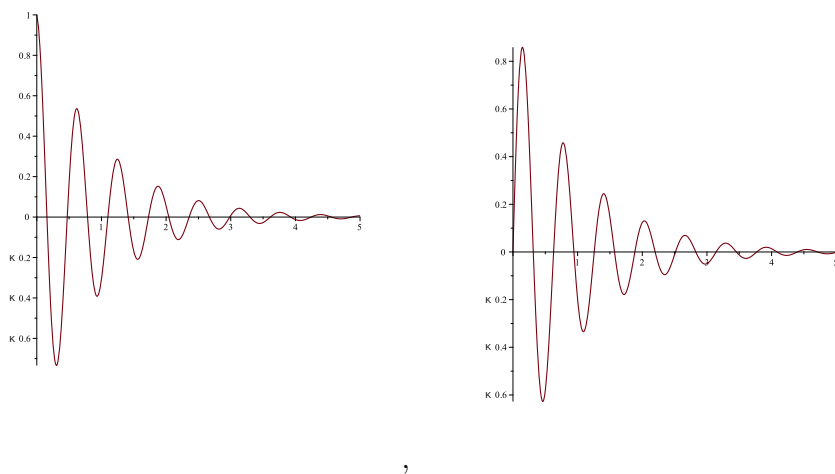


FIGURE 8. The graph of $e^{-t} \cos(10t)$ on the left and the graph of $e^{-t} \sin(10t)$ on the right $0 \leq t \leq 5$. The amplitude of the oscillations get smaller with time.

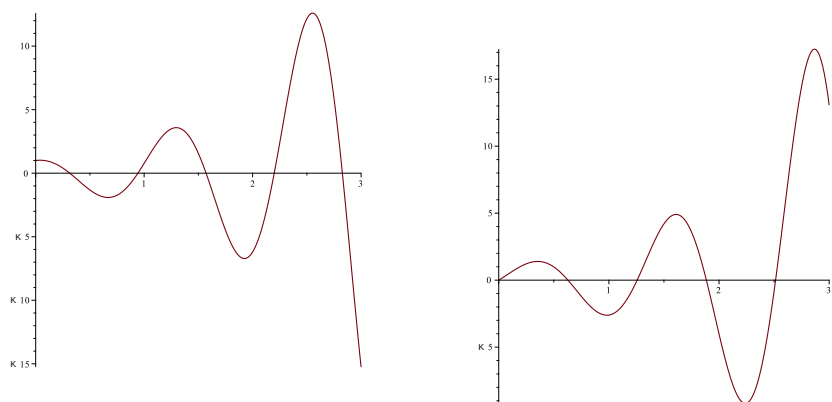


FIGURE 9. The graph of $e^t \cos(5t)$ on the left and the one of $e^t \sin(5t)$ on the right $0 \leq t \leq 5$. The amplitude of the oscillations get larger with time.